QP CODE: 25900869

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) REGULAR EXAMINATION October 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) –MG1DSCHPY100

Basics of physiology

(2024 ADMISSION ONWARDS)

Duration: 2Hours Maximum Marks: 70

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple ChoiceQuestions
Answer all questions

Each question carries 1 mark

1. d) Nervous tissue
2. b) CI
3. a) Ribs and sternum
4. c) Hypothalamus
5. d) Transcellular fluid
6. c) Lacuna

Part B
Short Answer Questions
Answer any **2** questions.
Each question carries **2** marks.

7. **Describe the structure and function of adherens junction**

- **Cell-to-cell connections** that link actin filaments through **cadherins** and lie just below tight junctions in epithelial tissues. (1 Mark)
- **Provide strong mechanical support,** helping tissues like **cardiac muscle and epidermis** withstand stress. (1 mark)

8. **Illustrate the functions of RBC**
   - transport of oxygen
   - transport of CO2

- buffering action
- blood group determination
- formation of bile pigments **(any four points – ½ mark each)**

9. Classify muscles depending on striations

- **Striated Muscle -** muscle which has a large number of cross-striations (transverse lines). Skeletal muscle and cardiac muscle belong to this category.
- **Non-striated Muscle-** Muscle which does not have cross-striations is called non-striated muscle. It is also called plain muscle or smooth muscle. It is found in the wall of the visceral organs. **(1 mark each)**

10. Define short bones

- Cube-like in shape, being approximately equal in length, width, and thickness. (**1 mark)**
- Short bones provide stability and support as well as some limited motion. Eg: carpals of the wrists and the tarsals of the ankles. (**1 mark)**

Part C

Short Essay

Answer any **4** questions

Each question carries **5** marks.

11. Describe the structure and functions of glandular epithelium. How are glands classified?

- Formed by **modified cuboidal or columnar epithelial cells** that become specialized for **secretion**. (1 marks)
- Based on number of cells: **Unicellular glands** – consist of single glandular cells. e.g., goblet cells of the alimentary canal.
  Multicellular glands – consist of clusters of cells. e.g., salivary glands. (2 marks)
- Based on presence or absence of ducts:
  Exocrine glands – have ducts and pour their secretions onto surfaces or into cavities. e.g., sweat glands, salivary glands.
  Endocrine glands – ductless glands that release hormones directly into the bloodstream. e.g., pituitary, thyroid. (2 marks)

12. **Describe special type of passive transport**

- **Passive transport** moves substances across the cell membrane without energy, occurring along the concentration gradient (from higher to lower concentration) to help maintain homeostasis (2 marks)
- **Bulk flow** is the movement of large quantities of substances from high pressure to low pressure, such as the exchange of oxygen and carbon dioxide across the respiratory membrane. (1 marks)

- **Filtration** is the movement of water and solutes from high hydrostatic pressure to low hydrostatic pressure, seen in capillaries and kidney glomeruli**.** (1 marks)
- **Osmosis** is the movement of water across a semipermeable membrane from lower solute concentration to higher solute concentration. **Types of osmosis include** endosmosis **(water moving into the cell) and** exosmosis **(water moving out of the cell).** (1 marks)

13. Classify oedema

- **Generalized oedema** - Oedema that involves the entire body **(1 mark)**
- **Local oedema** is the one that occurs is specific areas of the body such as abdomen, lungs and extremities like feet, ankles and legs. **(1 mark)**
- **Intracellular oedema** - the accumulation of fluid inside the cell. It occurs because of three reasons: a) Malnutrition, b) Poor metabolism , c) Inflammation of the tissues **(1 ½ mark)**
- **Extracellular oedema** is defined as the accumulation of fluid outside the cell. Causes for extracellular oedema are

  a) Abnormal leakage of fluid from capillaries into interstitial space.
- b) Obstruction of lymphatic vessels that prevents fluid return from interstitium to blood. **(1 ½ mark)**

14. Explain the composition and formation of lymph

- composition - water 96% and Solids 4%

solids

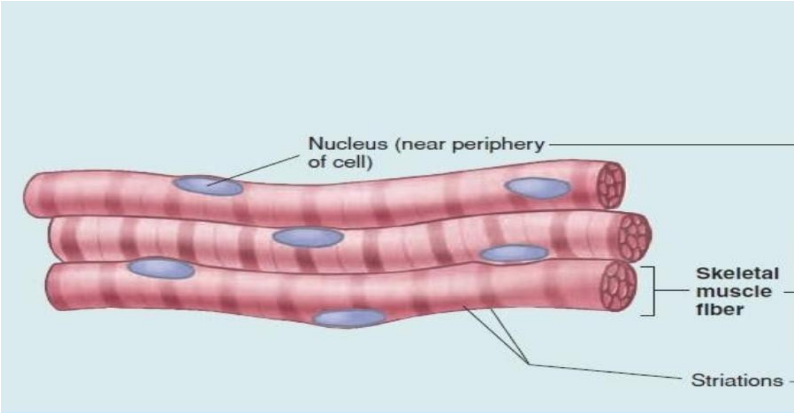
- Protein: Total protein content is roughly half that of plasma and varies from 2 to 4.5%.
- Fats: In fasting condition fat content is low but after a fatty diet it may be 5-15%.
- Carbohydrates (132.2 mg per 100 ml)
- Other constituents: Urea, non-protein nitrogenous substance, creatinine, chlorides, total phosphorus, inorganic phosphorus and calcium, enzymes and antibodies. **(2 marks)**
- **Formation** - leakage of fluid from capillaries to tissue spaces. Interstitial fluid that enters the lymphatic system.
- In humans, 20 liters of plasma is released into the interstitial space of the tissues each day due to capillary filtration.
- Out of the 20 litres, 17 litres is reabsorbed directly by the blood vessels and the remaining fluid seeps through the capillaries into the tissue spaces.
- Lymphatic system drains the excess fluid and empties it back into the bloodstream via a series of vessels, trunks, and ducts.

- Once this filtrate is out of the bloodstream and entered into tissue spaces, it is referred to as tissue fluid or interstitial fluid. Lymph is the term used to describe interstitial fluid once it has entered the lymphatic system. **(3 marks)**

15. Describe skeletal muscles with diagram

Diagram – 2 marks , Structure – 8 marks

Skeletal muscle is situated in association with bones forming the skeletal system. The skeletal muscles forms 40% to 50% of body mass and are voluntary and striated. They are supplied with somatic nerve fibres.



- Striated muscles are formed of large number of individual muscle cells known as muscle fibres. They are multinucleated and cylindrical in shape. The fibres do not branch in skeletal muscle.
- Each muscle fibre is covered completely by a thin layer of plasma membrane called the sarcolemma. It encloses a mass of protoplasm known as sarcoplasm. Sarcoplasm contains several nuclei, numerous mitochondria and a network of smooth endoplasmic reticulum called sarcoplasmic reticulum.
- Each muscle fibre is formed of large number of longitudinal myofibrils. Each of them is formed of delicate myofilaments which are arranged in repeating bundles or compartments called sarcomeres. This arrangement gives the muscle fibre a banded or striated appearance. Sarcomeres are structural unit of muscle fibres.
- Each myofibril shows dark and light bands. The dark band has a high refractive index and is called Anisotropic or A band. The light band is of low refractive index and is called Isotropic or I band. In the middle of the light band is the Z line or Z membrane. A sarcomere extends from one Z membrane to another Z membrane. The middle of the A band is occupied by a region of lighter colour known as H zone.

- Myofilaments are of two types – Thick and Thin. They are formed of proteins. Thick filaments are formed of protein myosin and are also known as myosin filaments. Thin filaments are formed by the protein actin and are also known actin filaments. In the peripheral darker portion of A band both actin and myosin filaments are present. In the I band only actin filaments are present where as in the H Zone only myosin filaments are present.

16. Explain the bones of pectoral and pelvic girdle

- Pectoral girdle - clavicle, scapula, **(2 marks)**
- Pelvic girdle

  Upper limb - humerus, ulna radius, carpals, metacarpals, phalanges. **(1 ½ marks)**

  Lower limb - femur, tibia, fibula, patella, tarsals, metatarsals, phalanges **(1 ½ marks)**

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

17. Explain cell signaling by describing its stages, types and role of signaling molecules.

- Cell signaling (**cell-to-cell or intercellular communicatio**n) is the process through which cells send and receive signals using hormones and other molecules to coordinate activities, respond to the environment, and maintain homeostasis. **(1 marks**)

- Three Stages of Cell Signaling:

Reception: A signaling molecule (ligand) binds to a receptor on the target cell.
Transduction: The signal triggers a cascade of biochemical reactions inside the cell.
Response: The cell produces a specific response, such as gene expression or enzyme activation.(**2marks**)

- Types

1. **Paracrine Signaling:**

Short-distance signaling where molecules released by one cell affect nearby cells. Responses are quick and short-lived.
Example: Neurotransmitters acting between adjacent neurons.

2. **Endocrine Signaling:**

Long-distance signaling where endocrine glands release hormones into the bloodstream to reach distant target cells. Responses are slower but long-lasting.
Example: Insulin from the pancreas regulating blood glucose.

3. **Autocrine Signaling**:

A cell releases molecules that bind to receptors on its own surface, producing a response in the same cell.
Example: Cancer cells often use autocrine signaling to stimulate their own growth. (**5Marks**)

- Types of Signaling Molecules:

Hormones (e.g., insulin, TSH) – chemical messengers carried in blood.
Neurotransmitters (e.g., serotonin, acetylcholine) – transmit signals across synapses.
Growth factors (e.g., EGF) – regulate cell growth, division, and differentiation
(2 Marks)

18. Explain structure and functions of lymphatic system
Structure (7 marks)

- Lymphatic system arises from tissue spaces as a meshwork of delicate vessels. These vessels are called lymph capillaries.
- Several lymph capillaries combine to form lymph vessels. All the lymph vessels join to form two large ducts, left and right lymphatic duct which brings lymph from right and left part of the body respectively. T
- he left lymphatic duct is also known as thoracic duct. The right lymphatic duct opens into the right subclavian vein and thoracic duct opens into the left subclavian vein. Thus it reaches the circulatory system.

Lymphatic Capillaries

- Lymphatic capillaries are vessels where interstitial fluid enters the lymphatic system to become lymph fluid. Located in almost every tissue in the body, these vessels are interlaced among the arterioles and venules of the circulatory system in the soft connective tissues of the body.
- Lymphatic capillaries are formed by a one cell-thick layer of endothelial cells and represent the open end of the system, allowing interstitial fluid to flow into them via overlapping cells. When interstitial pressure is low, the endothelial flaps close to prevent backflow. As interstitial pressure increases, the spaces between the cells open up, allowing the fluid to enter.
- In the small intestine, lymphatic capillaries called lacteals are critical for the transport of dietary lipids and lipid-soluble vitamins to the bloodstream. In the small intestine, dietary triglycerides combine with other lipids and proteins, and enter the lacteals to form a milky fluid called chyle. The chyle then travels through the lymphatic system, eventually entering the liver and then the bloodstream.
- **Lymphatic Vessels**

- Lymphatic vessels are the network of capillaries and large network of tubes located throughout the body that transport lymph away from tissues. Lymphatic vessels collect and filter lymph (at the nodes) as it continues to move toward larger vessels called collecting ducts. These vessels operate like our veins. They work under very low pressure, have a series of valves in them to keep the fluid moving in one direction.

Functions – (3 marks- ½ mark each for each point)

- It maintains fluid levels in our body tissues by removing all fluids that leak out of our blood vessels.
- It carries nutrients and oxygen to different parts of body.
- The lymphatic system is important for the optimal functioning of our general and specific immune responses.
- The lymph nodes monitor the lymph flowing into them and produce cells and antibodies which protect our body from infection and disease.
- The spleen and thymus are lymphatic organs that monitor the blood. They detect and respond to pathogens and malignant cells.
- The lymphatic system plays an important role in the absorption of fats from the intestine.

19. Explain physiological and pathological variations in WBC count

Physiological Variations in WBC Count

1. Diurnal variation: In morning the count is lowest. The count rises after midday and becomes highest in the evening.
2. Muscular exercise increases the count.
3. Adrenalin, emotional stress, fear, pain etc increases WBC count.
4. WBC count is about 20,000 per cu mm in infants and about 10,000 to 15,000 per cu mm of blood in children. In adults, it ranges between 4,000 and 11,000 per cu mm of blood.
5. Pregnancy-Higher during pregnancy being highest during parturition.
6. There is also an increase during menstruation.
7. Increased cellular destruction from infections produces derivative of nucleic acid and cause a rise in neutrophil.
8. Asthma, hay fever, skin diseases, parasitic infestation etc increases eosinophil count (eosinophilia).

9. ACTH causes an increase in neutrophil and decrease in lymphocytes and eosinophils.
10. Starvation and administration of chemicals like benzene produce leucopenia (increase in leucocytes).
11. Leucopenia also results due to bacterial, viral or protozoal infection, bone marrow disease or after exposure to ionizing radiations. **(1/2 mark each for each point)**

Pathological Variations

1. **Leukemia:** Leukemia is the condition which is characterized by abnormal and uncontrolled increase in leukocyte count more than 1,000,000/cu mm. It is also called blood cancer.
2. **Granulocytosis**: Increase of granulocytes in blood.
3. **Granulocytopenia**: Decrease of granulocytes in blood
4. **Agranulocytosis**: Great fall in the number of circulating granulocytes, which usually results from the harmful effect of certain drugs.
5. **Leucopenia**: Decrease in WBC count below 4000/cumm. Commonly this is due to fall in neutrophil cells. Certain drugs such as aminopyrin used for the relief of headache and cold are supposed to produce such a condition.
6. **Leucocytosis**: Rise in WBC count above 11,000/$mm^3$of blood. It is due to an increase in granulocytes. There might also be lymphocytosis and monocytosis. Internal bodily infections such as appendicitis may result in leucocytosis. In diseases like whooping cough and TB the lymphocytes may increase. In malaria and typhoid there is monocytosis.
7. **Neutrophilia**: Increase in number of neutrophils
8. **Eosinophilia**: Increase in the percentage of eosinophils. This condition occurs in tapeworm, hookworm infestations, skin diseases and in allergic conditions such as asthma.
9. **Lymphocytosis**: Increase in the number of lymphocytes. **(1/2 mark each for each point)**

20. Explain structural classification of joints

The structural classification of joints is based on whether the articulating surfaces of the adjacent bones are directly connected by fibrous connective tissue or cartilage, or whether the articulating surfaces contact each other within a fluid-filled joint cavity.

1) F**ibrous joint - (3 marks)**

At a fibrous joint, the adjacent bones are directly connected to each other by fibrous connective tissue, and thus the bones do not have a joint cavity between them.

a) **Suture -** It occurs only in the skull and possess short fibers of connective tissue that hold the skull bones tightly in place. No movement is possible here in adults.

b) **Syndesmosis -** the bones are more widely separated but are held together by a strap of fibrous connective tissue called a ligament or a wide sheet of connective tissue called an interosseous membrane. This type of fibrous joint is found between the shaft regions of the long bones in the forearm and in the leg.

c) **Gomphosis** - the narrow fibrous joint between the roots of a tooth and the bony socket in the jaw into which the tooth fits

2) Cartilaginous joint (2 marks)

The adjacent bones are united by cartilage, a tough but somewhat flexible type of connective tissue. These types of joints lack a joint cavity and involve bones that are joined together by either hyaline cartilage or fibrocartilage.

a) **Synchondrosis** - They are also known as primary cartilaginous joints. It is a cartilaginous joint where the bones are joined by hyaline cartilage, or where a bone is united to hyaline cartilage. It permits slight movement during early life and becomes rigid by the age of 20 years.

b) **Symphysis -** They are also known as secondary cartilaginous joints and are seen in axial bones. The bones are joined by fibrocartilage at these joints. They are slightly movable.

3) Synovial joint (5 marks)

The articulating surfaces of the bones are not directly connected instead they come into contact with each other within a joint cavity that is filled with a lubricating fluid.

a) **Pivot Joint**

It consists of the round ended end of one bone fitting into a ring formed by the other bone. This structure allows rotational movement as the rounded bone moves around its own axis. An example is the joint of first and second vertebrae of the neck that allows the head to move back and front.

b) **Hinge Joint**

In a hinge joint, the convex end of one bone articulates with the concave end of the adjoining bone. This type of joint allows only for bending and straightening motions along a single axis. Examples are elbow joint, the knee, ankle, and interphalangeal joints between the phalanges of the fingers and toes.

c) Condyloid Joint

They are also known as ellipsoidal joint. The shallow depression at the end of one bone articulates with a rounded structure from an adjacent bone or bones. Condyloid joints are biaxial joints that allow for two planes of movement.. Examples are knuckle joint, wrist joint etc.

d) Saddle Joint

At a saddle joint, both of the articulating surfaces for the bones have a saddle shape, which is concave in one direction and convex in the other. This allows the two bones to fit together like a rider sitting on a saddle. The primary example is the metacarpal bone at the base of the thumb.

e) Plane Joint

the articulating surfaces of the bones are flat or slightly curved and of approximately the same size, which allows the bones to slide against each other. The motion at this type of joint is usually small and tightly constrained by surrounding ligaments. Plane joints are found between the carpal bones of the wrist or tarsal bones of the foot.

f) Ball-and-Socket Joint

The joint with the greatest range of motion is the ball-and-socket joint. At these joints, the rounded head of one bone (the ball) fits into the concave articulation (the socket) of the adjacent bone. The hip joint and the shoulder joint are the only ball-and-socket joints of the body.